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PROJECT PHASE-I SYNOPSIS
ON

“DeepVision: AI-Based Intelligent Surveillance”

Submitted in the partial fulfilment of requirements for the award of Degree

Computer Science and Design

PROJECT ASSOCIATES

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PROJECT GUIDE

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CERTIFICATE

This is to certify that **Joycina Shiny M Achary, Lakshmi S Acharya, Arjun G R and Vaibhav S Wandkar** bearing USN: **4BD23CG015, 4BD23CG017, 4BD24CG401 and 4BD24CG409** respectively of Computer Science and Design department have satisfactorily submitted the report for the mini project entitled **“DeepVision: AI-Based Intelligent Surveillance”** in connection with 6th Semester **PROJECT (BCG685)**. The major project phase-1 has been approved, as it satisfies the academic requirements for the year 2025-26.

Project Guide :

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DeepVision: AI-Based Intelligent Surveillance

1. Title of the Project

DeepVision Surveillance: Detecting Suspicious Activities Using Artificial Intelligence

2. Introduction

The proposed project, **DeepVision Surveillance**, aims to develop an AI-powered surveillance system that can automatically monitor video feeds and detect suspicious behavior. The system will generate real-time alerts when abnormal activities are identified, enabling faster response and improving overall security efficiency.

Security and safety have become major concerns in today's world, especially in places such as airports, railway stations, shopping malls, banks, offices, and educational institutions. To address these concerns, surveillance systems using CCTV cameras are widely deployed to monitor activities and prevent potential threats.

Traditional surveillance systems mainly record video footage, which must be continuously monitored by human operators. However, with the rapid increase in the number of cameras, a huge amount of video data is generated every day. Manually analyzing this data is time-consuming, inefficient, and prone to human error, as security personnel may miss important events due to fatigue or lack of attention.

With the advancement of **Artificial Intelligence (AI)** and **Computer Vision**, it is now possible to develop intelligent systems that can automatically analyze video streams. These systems can detect objects, track movements, and identify unusual or suspicious activities in real time without constant human supervision.

By integrating AI with surveillance systems, this project transforms traditional passive monitoring into an active and intelligent system capable of enhancing safety and reducing human effort.

3. Problem Statement

DeepVision: AI-Based Intelligent Surveillance addresses the growing challenges faced by modern surveillance systems in handling and analyzing large volumes of video data.

Modern surveillance systems generate a massive amount of video data because cameras operate continuously across multiple locations. Monitoring this data manually is a challenging and inefficient task, as it requires constant attention from security personnel.

In traditional systems, CCTV cameras only record footage without actively analyzing it. As a result, identifying important events such as suspicious activities, unauthorized access, unattended objects, or unusual movements often requires reviewing long hours of video recordings. This process is time-consuming and may lead to delayed responses.

Additionally, human monitoring is prone to errors due to fatigue, distraction, or limited attention span. Critical events can be easily missed, which can compromise safety and security.

4. Objectives

The primary objective of this project is to develop an intelligent surveillance system that leverages Artificial Intelligence and Computer Vision techniques to automate the process of monitoring and detecting suspicious activities in real time.

The specific objectives of the project are:

- To design and develop an AI-powered surveillance system that can automatically analyze continuous video streams in real time.
- To implement object detection and tracking techniques to identify and monitor entities such as people, vehicles, and objects across frames.
- To study behavior and detect suspicious activities like unauthorized entry, abandoned objects, aggressive actions, and theft using smart algorithms.
- To generate real-time alerts with clear explanations to assist security personnel in making quick and informed decisions.

5. Methodology

The proposed system follows a systematic approach to automatically analyze video streams and detect suspicious activities using Artificial Intelligence and Computer Vision techniques. The methodology is divided into multiple stages, each responsible for a specific task in the overall process.

- **Video Input Acquisition**

The system captures video input either from a live CCTV camera, webcam, or a pre-recorded dataset. This serves as the primary source of data for analysis.

- **Frame Extraction and Preprocessing**

The input video is divided into individual frames for detailed analysis. Preprocessing techniques such as resizing, noise reduction, and normalization are applied to improve image quality and ensure consistent input for further processing.

- **Object Detection**

Each frame is analyzed using an object detection model (such as YOLO) to identify and classify objects like people, vehicles, and bags. This step helps in focusing only on relevant entities within the scene.

- **Feature Extraction and Behavior Analysis**

Important features such as speed, direction, and movement patterns of objects are extracted. These features are used to analyze normal behavior within the monitored environment.

- **Anomaly Detection**

A machine learning model is used to distinguish between normal and abnormal activities. Any deviation from learned patterns, such as unusual.

- **Alert Generation System**

When a suspicious activity is detected, the system generates real-time alerts. These alerts can be in the form of notifications, alarms, or messages to security personnel for immediate action.

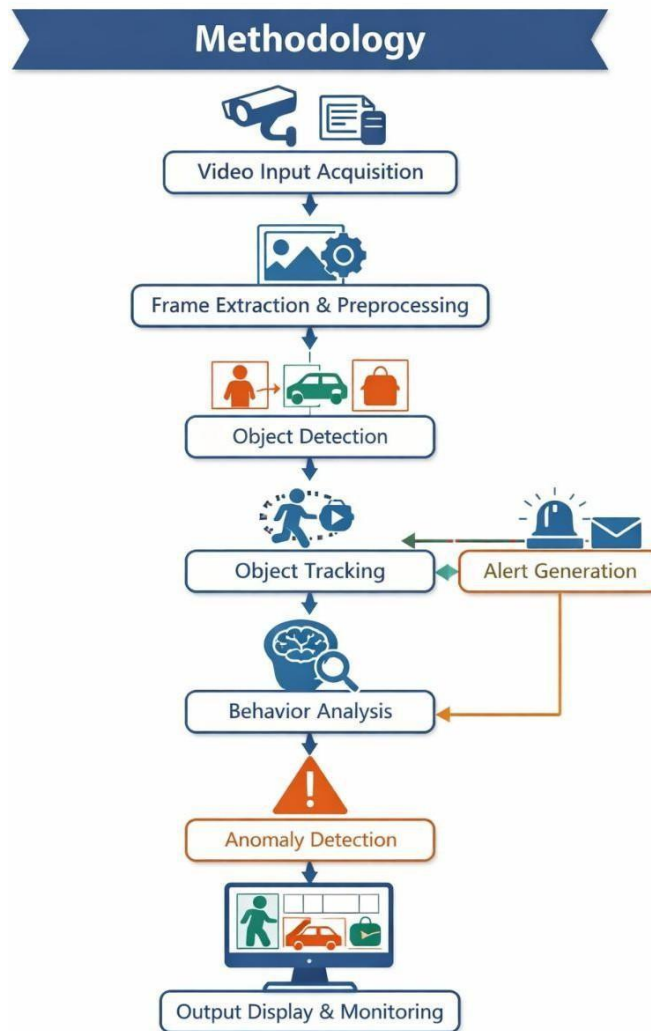


Fig: Methodology

6. Requirement Specification

The successful implementation of the proposed system requires both functional and non-functional requirements to be clearly defined. These requirements ensure that the system performs its intended tasks efficiently while maintaining reliability and performance.

Functional Requirements:

Functional requirements describe the core operations that the system must perform:

- The system shall capture video input from live surveillance cameras (webcam/CCTV) as well as pre-recorded video datasets for analysis.
- The system shall preprocess video frames to enhance quality and prepare them for accurate detection and analysis.

- The system shall implement object detection techniques to identify and classify entities such as people, vehicles, and unattended objects within each frame.
- The system shall incorporate object tracking mechanisms to continuously monitor detected entities across consecutive frames and maintain their identities.
- The system shall analyze movement patterns and behavioral characteristics of tracked objects to understand activity trends over time.
- The system shall detect suspicious activities such as loitering, unauthorized access, abandoned objects, physical altercations, abusive or aggressive behavior, and theft-related incidents including robbery.
- The system shall apply rule-based and machine learning-based approaches to distinguish between normal and abnormal activities in different scenarios.
- The system shall generate real-time alerts and notifications when suspicious activities are detected, ensuring immediate attention from security personnel.
- The system shall provide explainable outputs by indicating the reason behind each detected alert (e.g., prolonged presence, unusual movement patterns, restricted zone entry).
- The system shall allow basic configuration of parameters such as detection thresholds, time limits for loitering, and restricted zones.
- The system shall support logging and storage of detected events for future analysis and review.

Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system:

- The system shall support near real-time processing with minimal latency to ensure timely detection and response.
- The system shall maintain high accuracy in object detection, tracking, and activity recognition under different environmental conditions.
- The system shall be reliable and robust, capable of functioning continuously for long durations without failure.
- The system shall be scalable to support multiple camera inputs and larger surveillance environments.

- The system shall provide a user-friendly interface for monitoring video feeds, alerts, and system outputs.
- The system shall ensure data security by restricting unauthorized access to video feeds and system outputs.
- The system shall incorporate basic privacy-preserving mechanisms such as optional face blurring or anonymization where required.
- The system shall be maintainable and modular, allowing easy updates, improvements, and integration of new features.
- The system shall be compatible across different platforms and development environments.

7. Technologies Used

The proposed system uses the following technologies for implementation:

- **Programming Language**
 - Python
- **Computer Vision Library**
 - OpenCV
- **Machine Learning Frameworks**
 - TensorFlow
 - PyTorch
- **Object Detection Model**
 - YOLO (You Only Look Once)
- **Development Environment**
 - Visual Studio Code
- **Alert & Notification System**
 - SMTP (Email Alerts)
 - Twilio – Used for sending SMS notifications
- **Hardware Requirements**
 - Webcam or CCTV Camera
 - Computer System (CPU/GPU)
- **Operating System**
 - Windows
 - Linux

8. Expected Outcome:

The proposed system is expected to deliver an efficient and intelligent surveillance solution by automating the process of monitoring and detecting suspicious activities. It is expected that the system will effectively track detected objects across multiple frames and analyze their movement patterns to understand behavioral characteristics. Based on this analysis, the system will be able to identify suspicious activities such as loitering, unauthorized access, abandoned objects, physical altercations, and theft-related incidents including robbery. It will showcase the practical application of Artificial Intelligence and Computer Vision in enhancing surveillance systems. Furthermore, the system is expected to be scalable and adaptable, allowing it to be deployed in various environments such as campuses, offices, public areas, and transportation systems. Overall, the project will result in a functional prototype that highlights improved efficiency, reduced human effort, and enhanced security through intelligent video analysis.

The expected outcomes of the project are:

- The system will automatically analyze continuous video streams in real time without the need for constant human monitoring, reducing manual effort and improving efficiency.
- It will accurately detect and classify objects such as people, vehicles, and unattended items within the surveillance area using advanced object detection techniques.
- The system will track detected objects across frames and analyze their movement patterns to understand behavior over time.
- It will identify suspicious or abnormal activities such as loitering, unauthorized access, and abandoned objects based on predefined rules and behavioral analysis.
- The system will generate real-time alerts and notifications, along with brief explanations, to help security personnel quickly understand and respond to potential threats.
- Overall, the project will demonstrate how artificial intelligence can enhance traditional surveillance systems by making them more proactive, intelligent, and reliable.

9. Innovation and Significance:

The proposed project introduces innovation by integrating Artificial Intelligence and Computer Vision into traditional surveillance systems. Unlike conventional systems that only record video footage, this system actively analyzes video streams in real time and detects suspicious activities automatically.

The use of object detection and anomaly detection techniques allows the system to identify unusual patterns such as abnormal movement, unauthorized access, or unattended objects. This shifts surveillance from passive monitoring to an intelligent and proactive approach.

The significance of this project lies in its ability to improve security and reduce human dependency. By automating the monitoring process, it minimizes errors caused by human fatigue and increases the chances of detecting critical events on time.

This system has wide applications in areas such as public safety, smart cities, transportation systems, offices, and educational institutions. It can also contribute to further research and development in the field of AI-based surveillance and intelligent security systems.

Another significant feature of the project is the incorporation of explainable outputs. The system not only generates alerts but also provides clear reasons for why a particular activity is considered suspicious, such as prolonged presence in a restricted area or unusual movement patterns. This improves transparency and helps users better understand and trust the system's decisions.

The project also emphasizes real-time processing and automated alert generation. By instantly notifying security personnel about potential threats, the system reduces response time and enhances overall security efficiency. This shift from reactive monitoring to proactive detection represents a major improvement over traditional surveillance approach.

The importance of this project lies in its ability to improve security and reduce human dependency. By automating the monitoring process, it minimizes errors caused by human fatigue and increases the chances of detecting critical events on time.

10. Conclusion

In conclusion, the proposed system demonstrates how Artificial Intelligence can enhance traditional surveillance methods by enabling automatic detection of suspicious activities. By combining computer vision and machine learning techniques, the system can efficiently analyze video data, identify abnormal behavior, and generate real-time alerts. This reduces the need for continuous human monitoring and improves overall system efficiency. The project highlights the potential of intelligent surveillance systems in ensuring safety and security in modern environments and serves as a foundation for future advancements in AI-driven monitoring solutions. The integration of computer vision and machine learning techniques allows the system to not only detect events but also analyze behavioral patterns and generate meaningful alerts. This improves the efficiency and responsiveness of security systems, helping personnel take timely action.

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